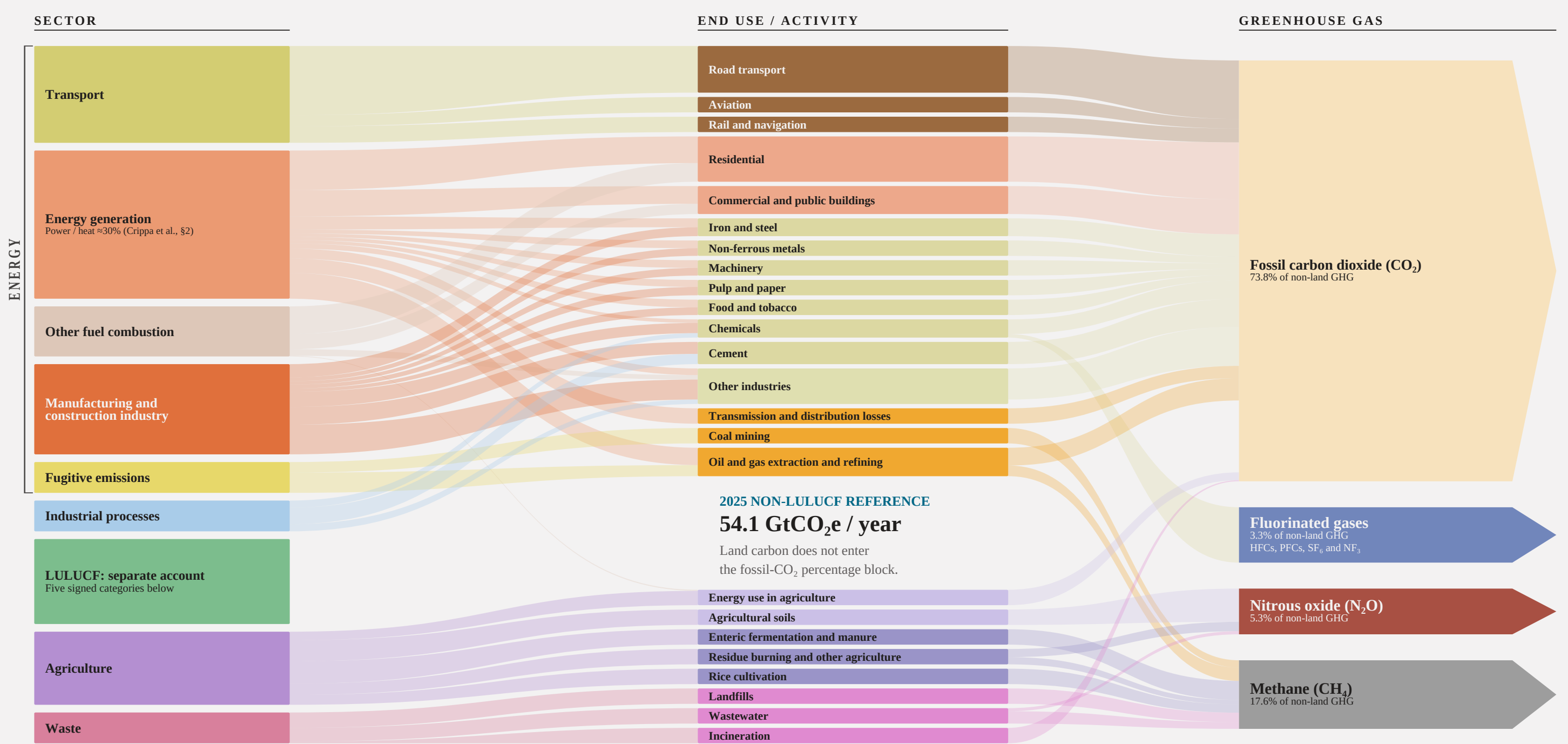
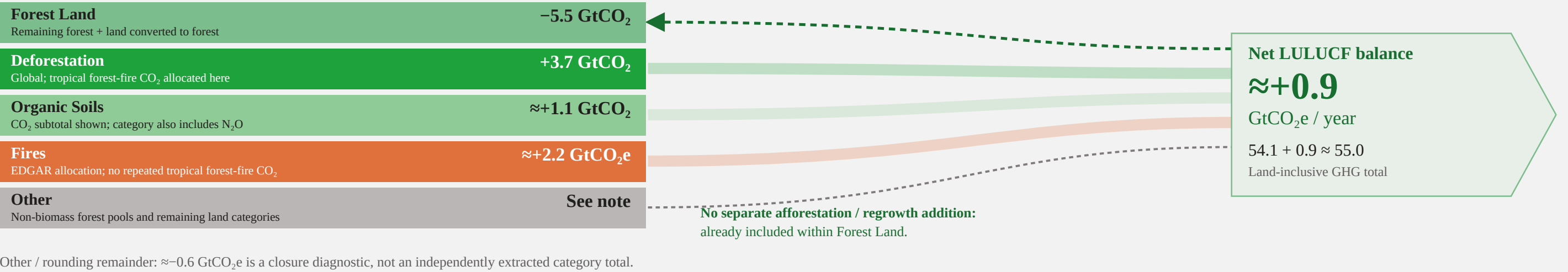




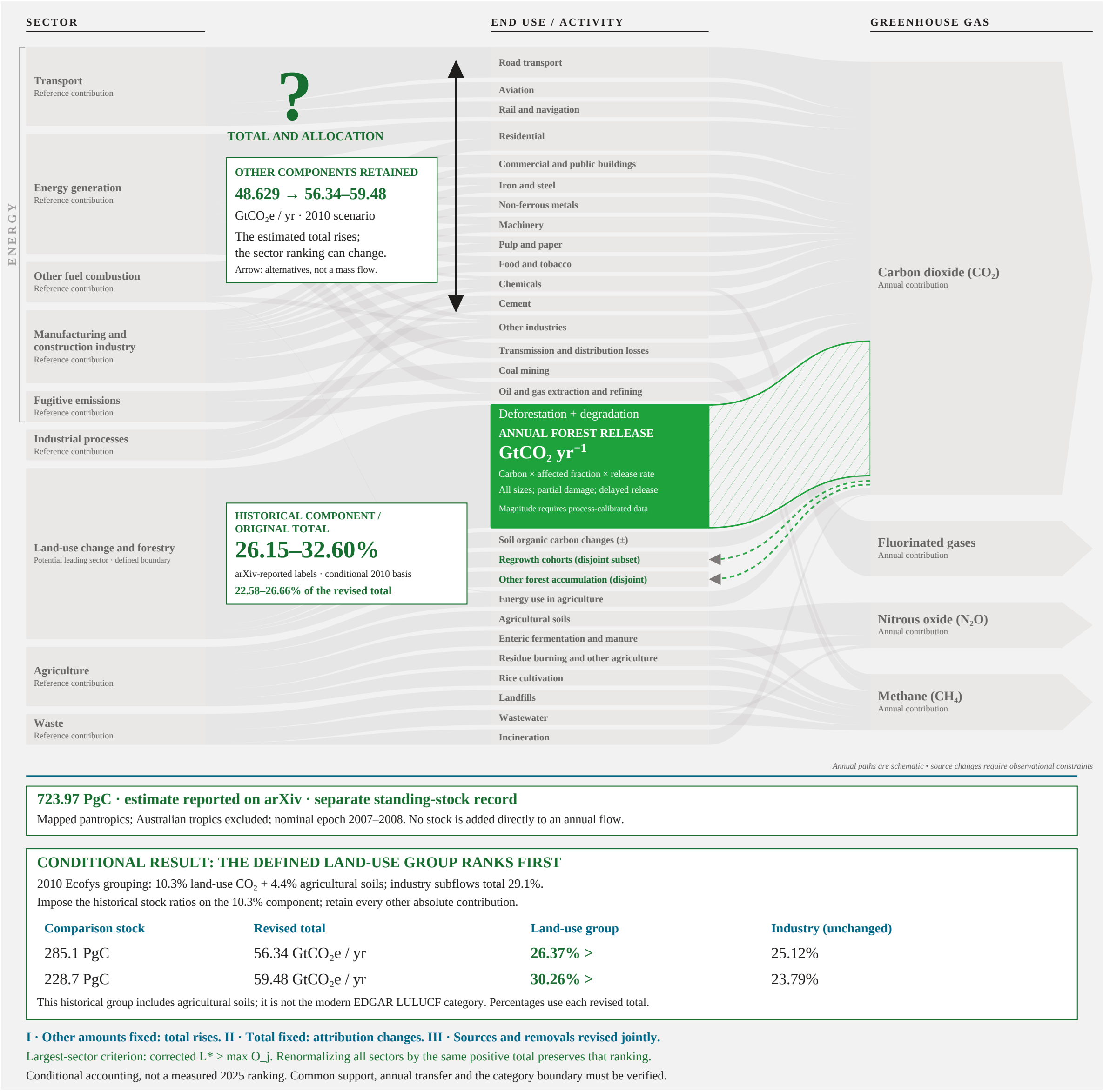
THE SIGNED LAND ACCOUNT

EDGAR 2026 · Section 3, Figure 6 and Annex 2 · 2025 estimates



Disjoint land categories, separate gas denominator. Forest Land already includes land converted to forest; no extra afforestation or regrowth subtotal is added. Deforestation includes tropical forest-fire CO₂, which is excluded from Fires; tropical fire non-CO₂ gases remain in Fires. Organic Soils’ quoted 1.1 is CO₂ only. The diagnostic remainder 0.9 − [3.7 − 5.5 + 2.2 + 1.1] ≈ −0.6 also absorbs omitted subcomponents and rounding; it is not a separately extracted Other estimate. Source: Crippa et al. (2026), booklet Sections 2–3 and Annex 2, doi:10.2760/7717504 (CC BY 4.0).

Figure 1: The reported 2025 account and its boundaries. The sector–activity–gas organization is adapted from WRI (2005) and Ecofys (2013), with author-created vector artwork and a separate five-category signed land account. Intermediate activity nodes describe pathways, rather than EDGAR’s official sector list. Gas shares refer only to EDGAR’s 54.1 GtCO₂e non-LULUCF inventory using AR5 GWP₁₀₀; 73.8% is fossil CO₂. Land emissions therefore do not enter that gas block. The reverse dashed link denotes Forest Land’s net removal. Forest/deforestation estimates use a global Tier 1 approach with default parameters and estimated activity data beyond the latest input maps, not independently remeasured 2025 carbon densities. Paths and all graphical areas are schematic, not a balanced quantitative Sankey. Electricity-attribution paths must not be counted again as direct emissions.



All stem sizes and explicit carbon fate. Per-tree carbon, abundance, affected fractions and release timing determine annual deforestation and degradation. An omitted positive net source leaves actual accumulation above the account’s reconstruction unless other changes offset it. Observed air can already contain emissions missed by a bottom-up inventory. Carbon released after clearing remains in its later-year balance; no unmeasured compensating sink is imposed.

Figure 2: The numerical second interpretation and its conditional largest-sector inference. The question mark and double-headed arrow concern the estimated total and sector allocation, not an established error in a named energy source. The historical annotation restores the arXiv-reported 26.15–32.60% range; nearest rounding from displayed inputs gives 26.16–32.61%. Under the imposed historical annual multiplier and retained other absolute contributions, totals become 56.34–59.48 GtCO₂e yr⁻¹. In Ecofys’s 2010 grouping, adding the retained 4.4% agricultural-soils component makes its land-use group exceed the 29.1% industry subflow sum at both endpoints, before and after renormalization. That grouping differs from modern EDGAR LULUCF and from the conceptual sector layout above; the numerical card is a separately defined historical example, not a measured 2025 allocation. The component alone ranks above industry only at the upper endpoint. Supplementary Note S3 states the transfer, source-provenance and denominator assumptions. The 723.97 PgC stock remains separate from the annual network. All widths and arrows are schematic; the energy supersector is not a single peer category in the ranking. Layout adapted from WRI (2005) and Ecofys (2013); source interpretation: Arellano-P. and Rangel-Ch. (2015/2016).

CARBON ACCURACY STARTS WITH THE REFERENCE

Spatial coverage measures where; a validated material conversion determines how much.

A • PROXY-LABEL PATHWAY

Reduced predictor measurements

Diameter, canopy height, nominal density and context.
Organ architecture may remain only implicitly represented.



Allometric biomass / carbon labels

A fitted response summarizes its calibration population.
Transport can fail when architecture or composition changes.



Remote-sensing calibration

LiDAR or optical metrics are fitted to reference labels.
Accurate prediction of those labels is not independent mass validation.



Expanded spatial prediction

More coverage can reduce sampling or mapping uncertainty.
A shared conditional carbon-scale bias can remain.

INHERITANCE IDENTITY • IDEAL CONDITIONAL PREDICTION

$E[\text{label} \mid \text{LiDAR}] - E[\text{true carbon} \mid \text{LiDAR}] = E[\text{reference-label bias} \mid \text{LiDAR}]$

Conditional mean-zero residual noise is assumed. This is a mechanism, not an empirical global correction factor.

Initial estimate reported on arXiv

723.97 PgC · architecture-informed calibration
Nominal map epoch 2007–2008; Australian tropics excluded.
221 reconstructed trees; 2,232 tissue samples reported.
Successor calibration in development; no new total reported.

B • ARCHITECTURE-INFORMED PATHWAY

Measured organs and material

Stems, branches, bark and other explicit compartments.
All stem sizes; biological volume, not saleable timber only.



Carbon from geometry and tissue

Carbon = Σ (organ volume × basic density × carbon fraction).
Direct laser reconstruction may help measure woody volume.



Independent calibration and holdouts

Validate material geometry, density, chemistry and sampling.
Keep reference sites separate from model fitting.



Spatial extension with traceable targets

LiDAR can extend a separately validated carbon reference.
Check transfer, omitted structures and regional extrapolation.

From carbon measurement to annual accounts

All sizes × abundance × affected fractions × release histories.
Deforestation, degradation, regrowth and other removals.
Uncounted net release raises reconstructed accumulation.
A retained total constrains reattribution among sources.

NOT EVERY LiDAR PRODUCT USES THE SAME MEASUREMENT CHAIN

Disney et al. (2018); Calders et al. (2022); Wilkes et al. (2023): independent TLS reconstruction pathways.

Measure architecture and organ allocation to identify the source of allometric calibration error.

Refining calibration can improve the stock estimate without changing the distinction between reference accuracy and coverage.

Sources: Arellano-P. & Rangel-Ch. (2016); Goodman et al. (2014); Calders et al. (2015); GEDI L4A V2 documentation.

Operational diagram: no graphical width represents a measured carbon share. Henry Arellano-Peña.

The calibration distinction. A remote-sensing product fitted to allometric reference labels can inherit their conditional bias. Direct reconstruction of biological material with measured tissue properties provides a different reference pathway, which can itself use terrestrial laser measurements. Independent reference tests are required for both approaches; greater coverage alone does not determine absolute carbon accuracy.

Figure 3: Two measurement-to-map pathways and their annual-accounting consequences. The left pathway extends a calibrated proxy; the right begins with organ geometry and matched material properties. Architecture and tissue variation identify mechanisms by which an allometric predictor can depart from actual biological carbon. The conditional-expectation identity assumes a source distribution and mean-zero residual noise conditional on the sensor variables; a real learner also has fitting and transfer errors. The initial reconstruction and physical tests are reported in the arXiv study of Arellano-P. and Rangel-Ch. (2015/2016). Crown-related biomass discrepancies are independently illustrated by Goodman et al. (2014). Disney et al. (2018), Calders et al. (2015, 2022) and Wilkes et al. (2023) document terrestrial laser material reconstruction and segmentation, while the GEDI L4A Version 2 guide documents a model linking waveform height metrics and biomass reference data. These examples retain their study domains and establish no universal multiplier. Total material volume includes nonmerchantable woody organs; belowground and dead pools remain separately defined. The 723.97 PgC value is the initial estimate reported on arXiv. Its annual implications require affected populations, release histories and removals. Neither larger spatial coverage nor numerical reproduction of calibration labels is equivalent to independent carbon-mass validation. Original diagram: Henry Arellano-Peña.